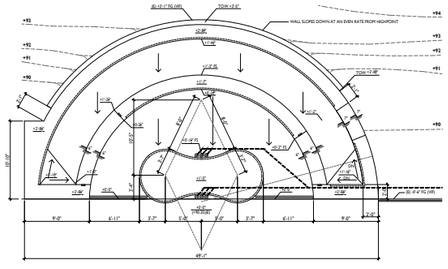
West Edge Architects Work

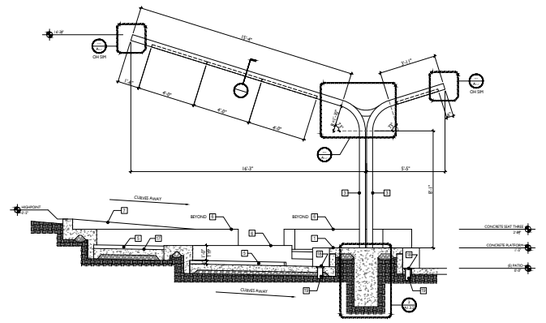
Presentation Room and Outdoor Area at a College

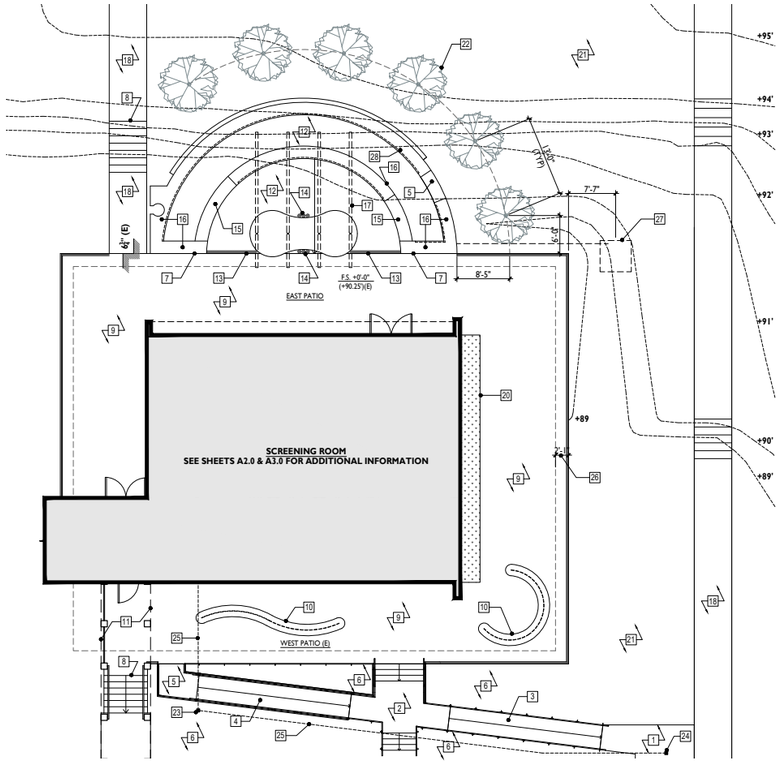


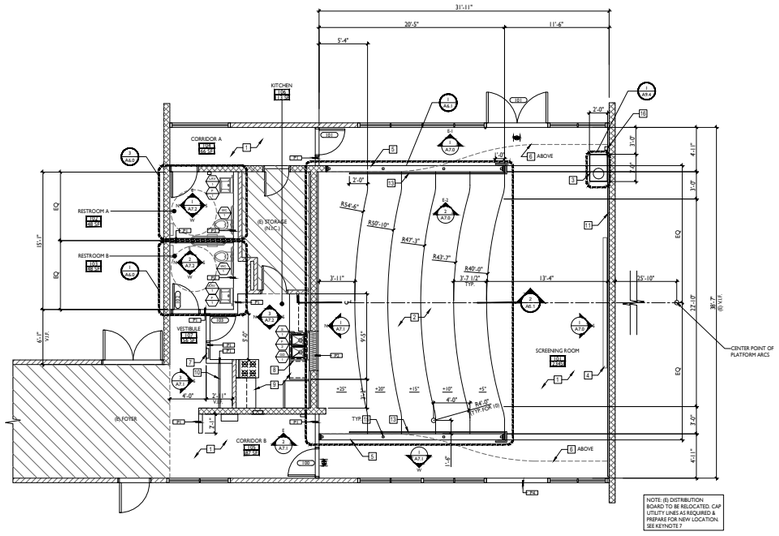




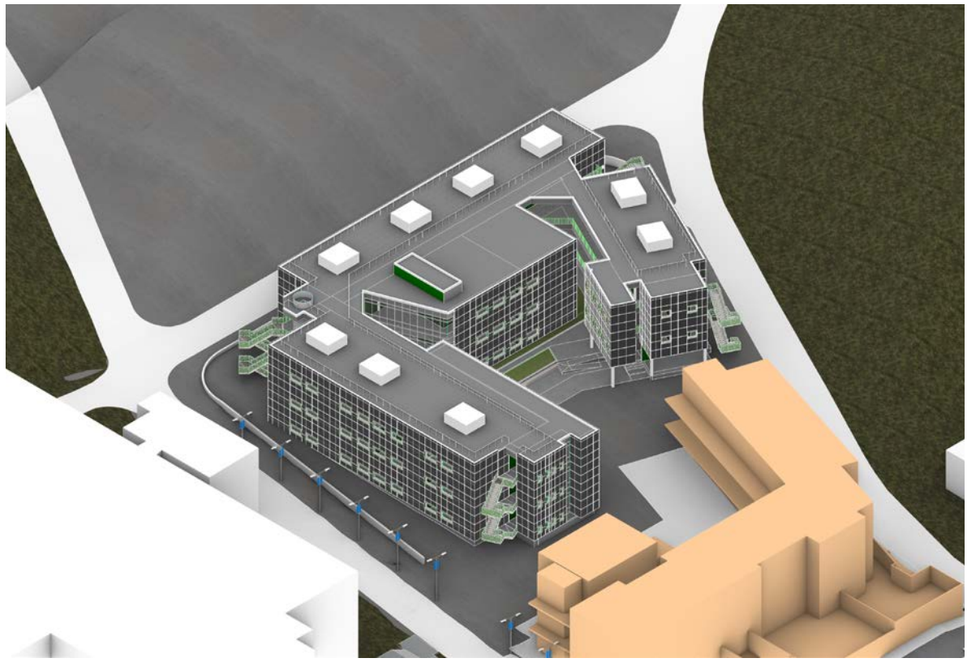




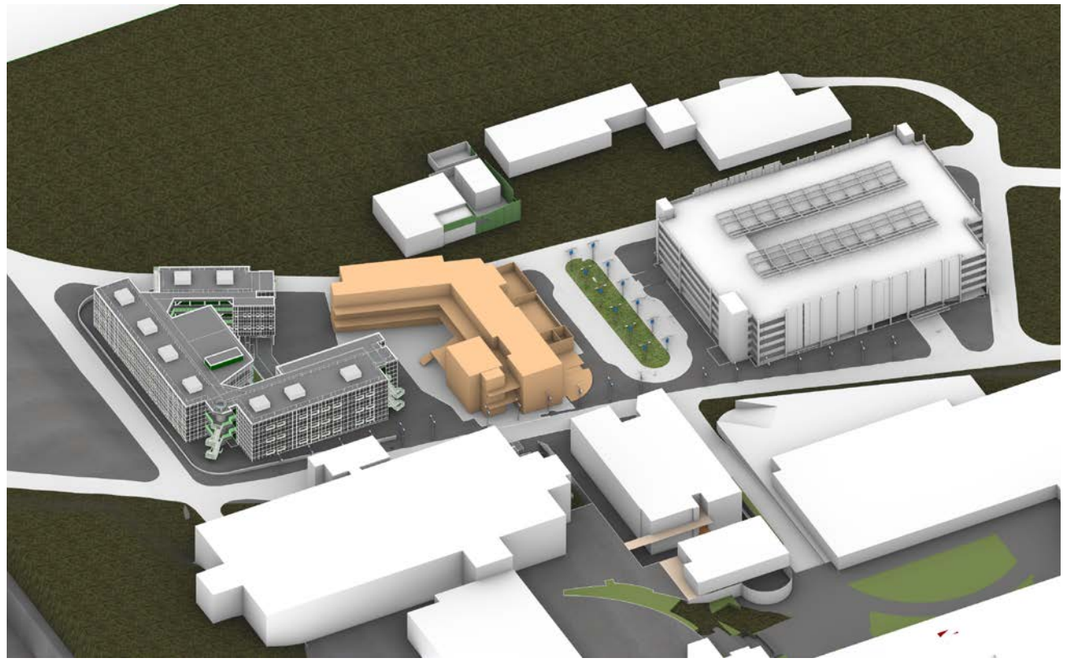




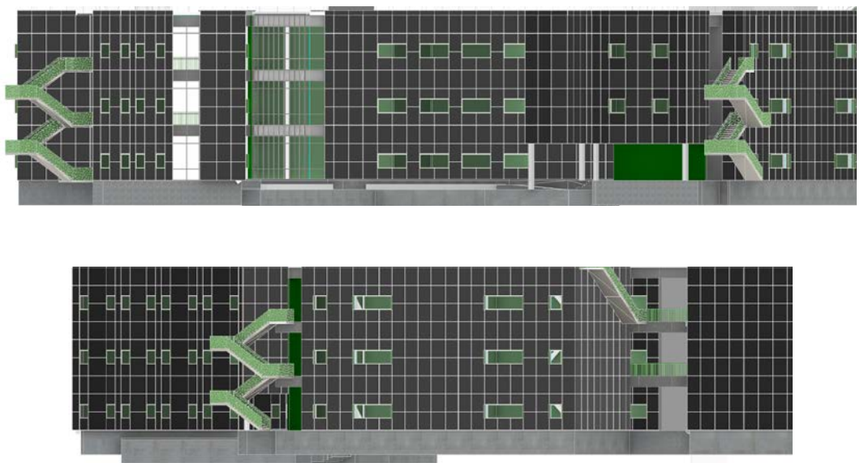
Science Building at a College



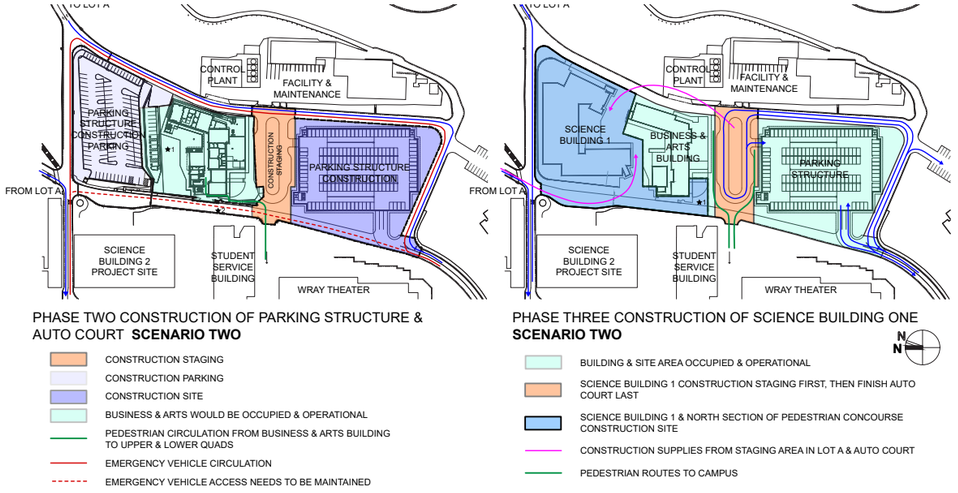
College Building Model



Site Model



Elevation Study



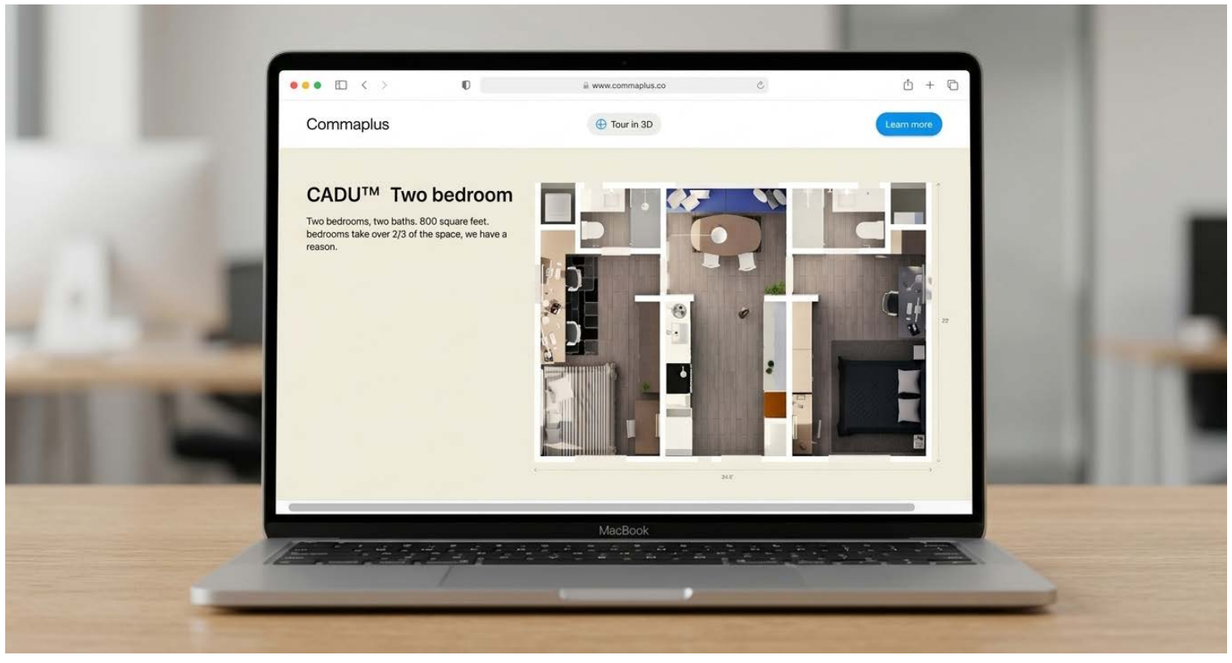
Site Diagram

Commaplus Work









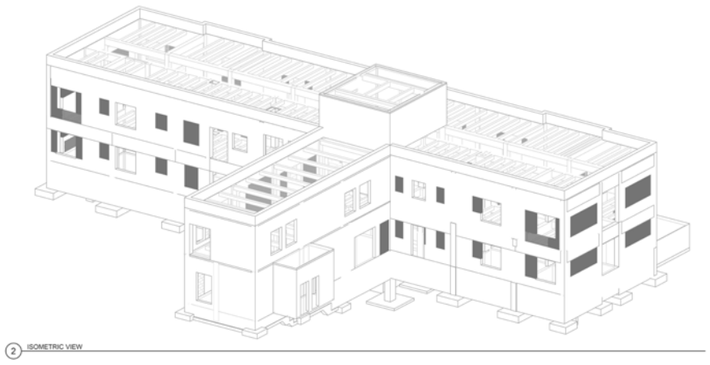




ZWA Work

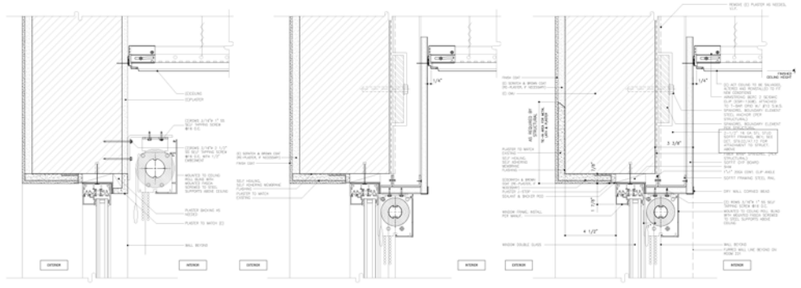


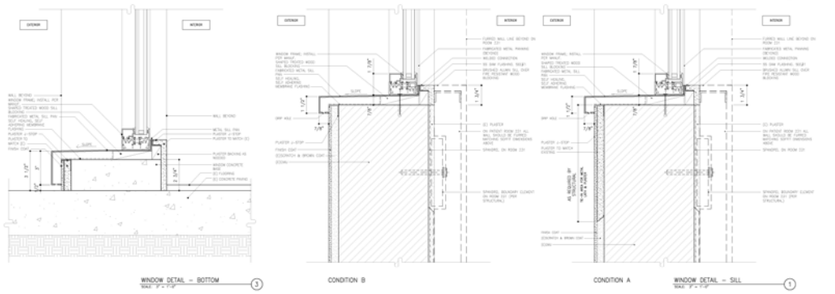
1 ISOMETRIC VIEW

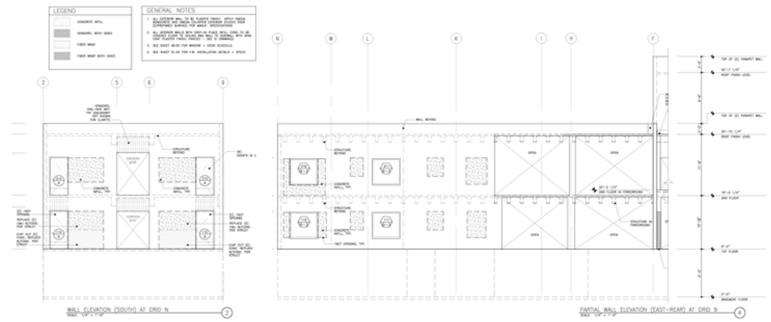


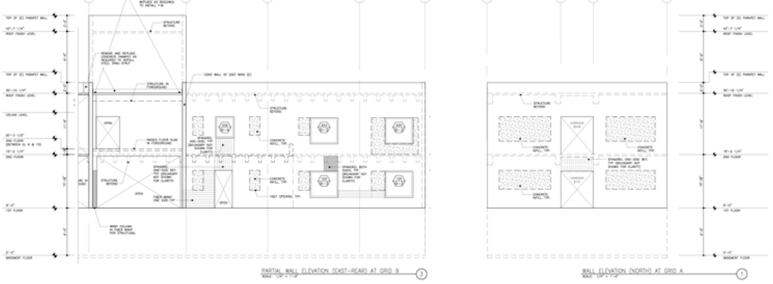
2 ISOMETRIC VIEW











Hyperloop Work







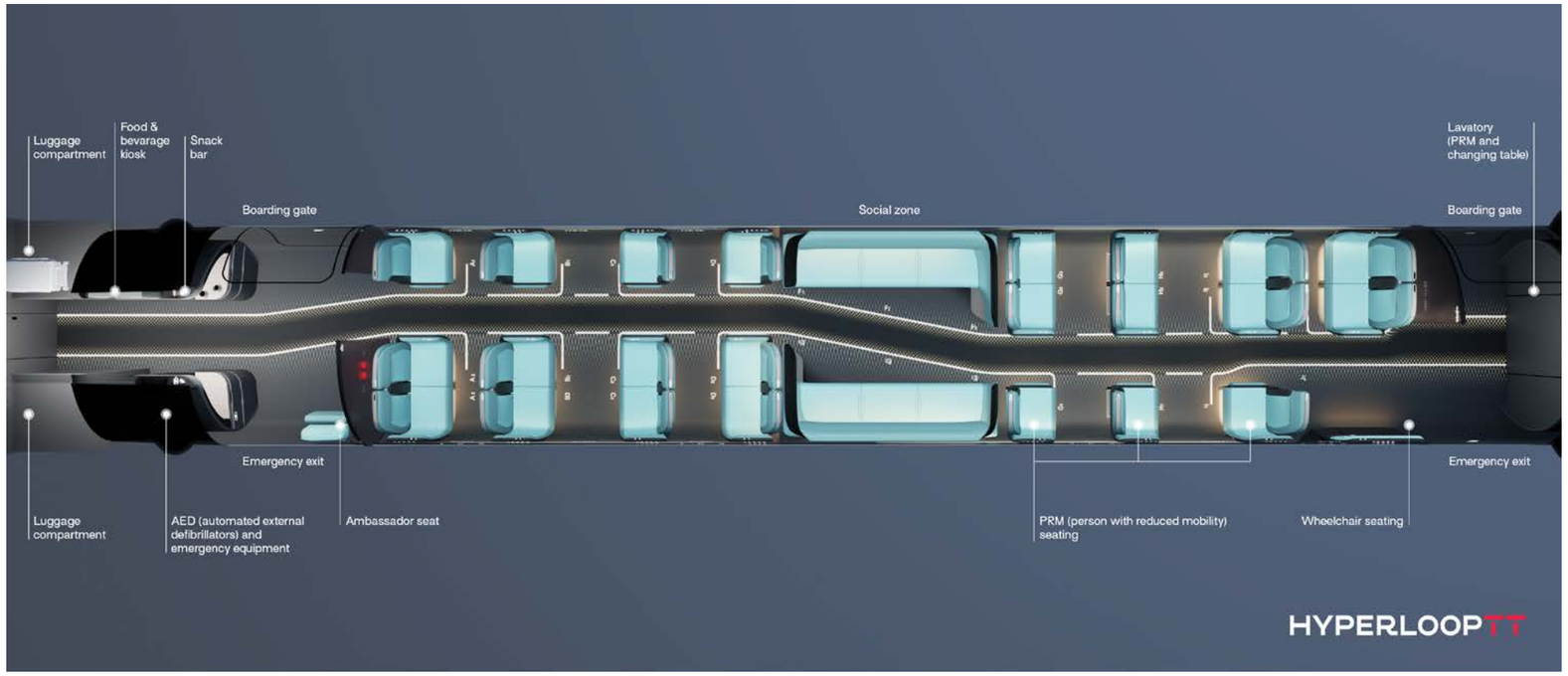


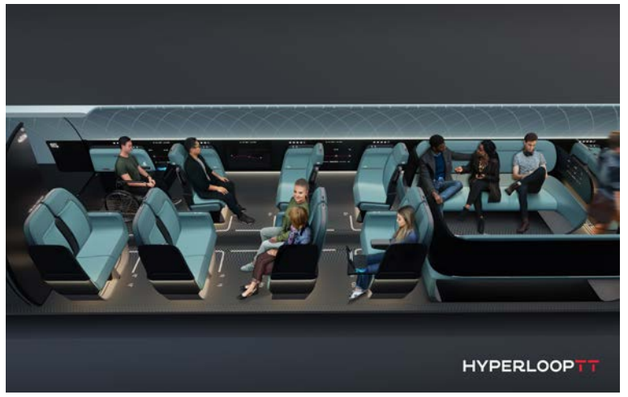


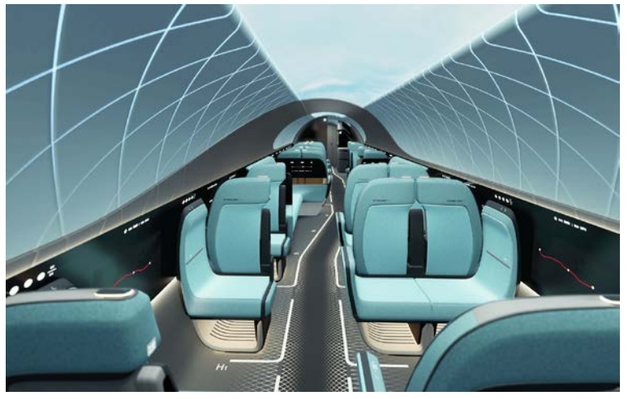


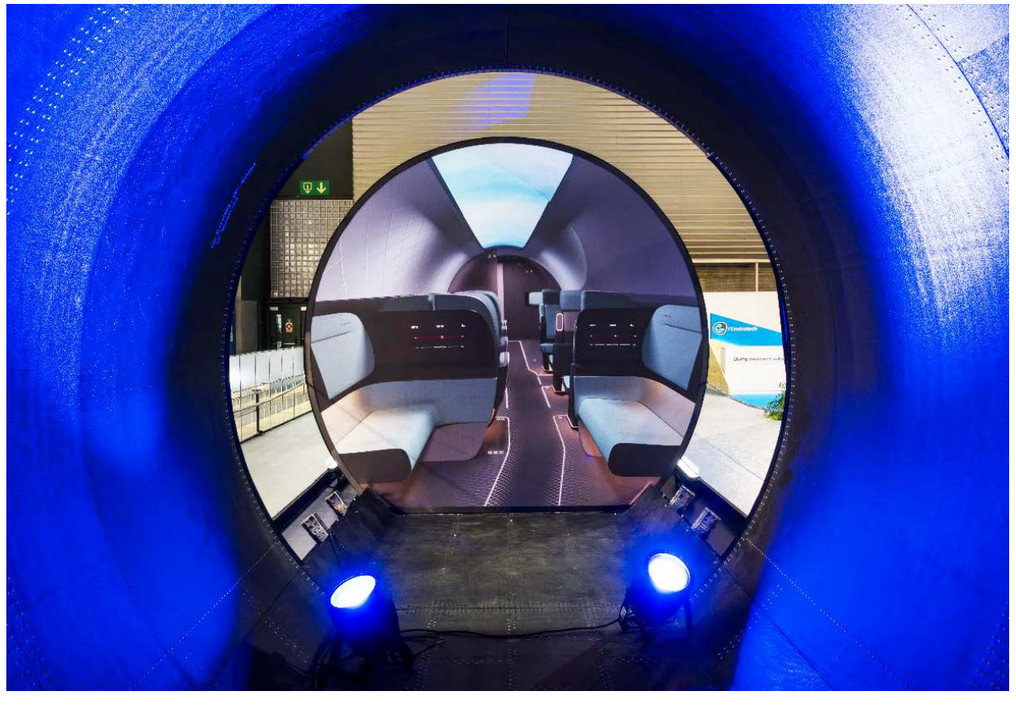





Luggage compartment
Food & bevarage kiosk
Snack bar
Boarding gate
Social zone
Lavatory (PRM and changing table)
Boarding gate
Emergency exit
Emergency exit
Luggage compartment
AED (automated external defibrillators) and emergency equipment
Ambassador seat
PRM (person with reduced mobility) seating
Wheelchair seating
HYPERLOOPTT


HYPERLOOPTT










WELCOME EMMA
HYPERLOOPTT

Urban Scale Computation
The Mega Data Center of Future

UCLA Architectural Studio
Thesis Project
Masterplan Design

2021.9-2022.6

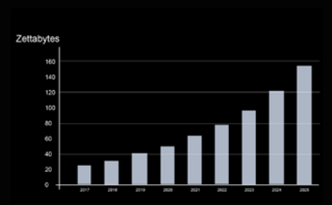
Site: Orkney islands, Scotland



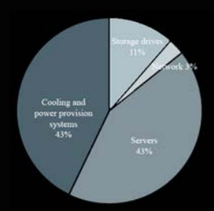
This project imagines a data center at the scale of a city, built for the world's insatiable need to store and process data. The system is designed to grow and maintain itself, so no one has to work in its hazardous conditions. It is also a test field: what happens when design decisions are handed to AI instead of human judgment?



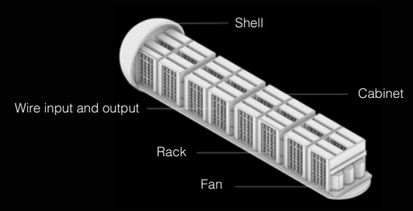
The yellow square shows how much area will the servers take if spreaded evenly.



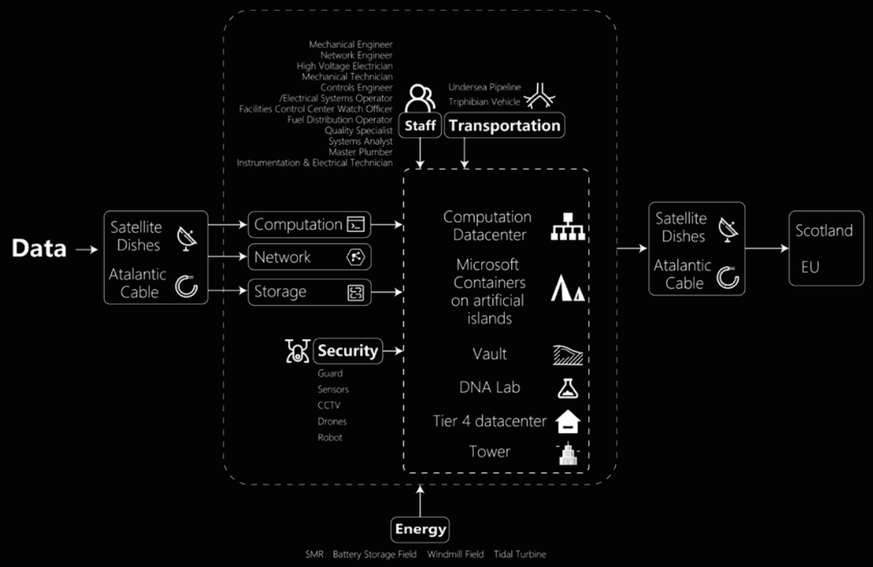
Cloud storage and computation has become one of hottest business due to the extremely expanding global data size.The cloud storage data size doubles every three years. It is top priority to build a growable mega data center.



Data centers are notorious energy consumers, especially at this large scale. It is shown that cooling takes up 43% of the energy used. If the energy of cooling is saved, the total energy consumption will reduce to half, saving billions of pounds.



Microsoft has successfully put a capsule containing servers in the cold water of the North Sea, which inspired us to put data in sea water as well. However, millions of times of what Microsoft did.





The input and output towers transmit data through the Atlantic cables and satelite dishes. They are located at the northest point of the project according to the location of the undersea cables.



The mophology of the project is inspired from the machine learning module: CycleGan. Generally, the artifical islands, in the form of piers, form complex clusters that look like circuit board. The huge data towers stand in between the clusters.

A cluster a basic unit of the system. In one cluster, multiple towers of different shapes and programs work with each other. The wind turbines and tidal turbines generate power and store excessive energy in the battery that are on the piers.

The orange line is the main piers. They extend from south to north and act as the artery of the project. Some minor piers grow from the main piers.
In spite of the linear shape, they are actually artifical islands.

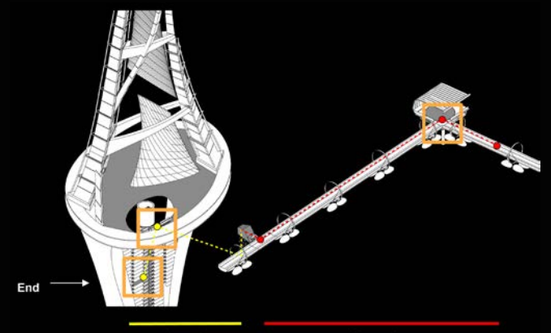


The mega factory have a length of a few miles. It is a superstructure that manufacture almost everything required in the project. Its large scale guarantee the possibility of such a urban scale data center.

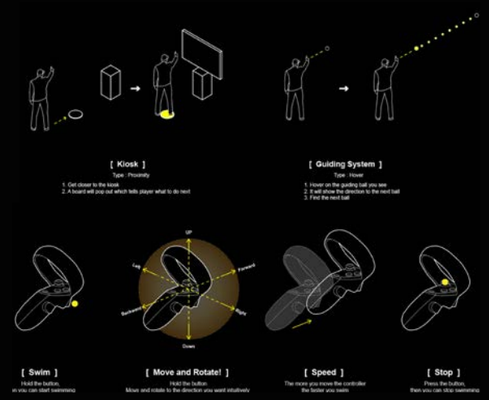


Westray Island
Eday Island
Rousay Island
Mainland
Shapinsay Island
Kirkwall
Hoy Island

- Satellite Dishes
- Atalantic Cable
- Security
- SMR
- Battery Storage Field
- Windmill Field
- Tidal Turbine
- Satellite Dishes
- Atalantic Cable
- Vault
- DNA Lab
- Tier 4 datacenter
- Tower
- Computation Datacenter
- Microsoft Containers on artificial islands
- Computation
- Network
- Storage
- Transportation
  Triphibian Vehicle
  Undersea Pipeline
- Residential

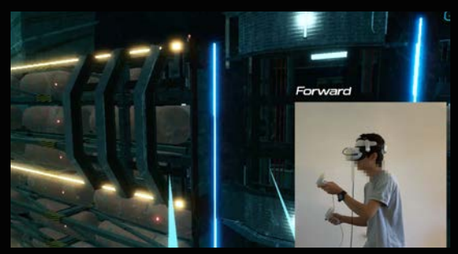


The VR experience is complementary to the overall display of the project and will present more details of the undersea world on human scale. The player will act as a diver to fix a few bugs in the server racks. We will see the hologram of a part of the project. We can walk on the inconventional piers. We can have a better understanding of the hiearchy of the very complex system of the whole project.



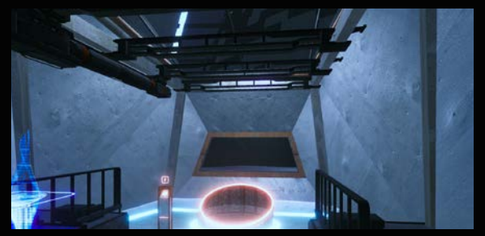
We create a brand new experience of VR. Very few games can simulate the experience of diving in VR. We create diving motion with very much freedom and a immersive, realistic underwater world.

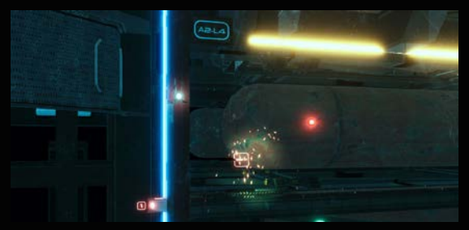
The VR gameplay, made with Unity









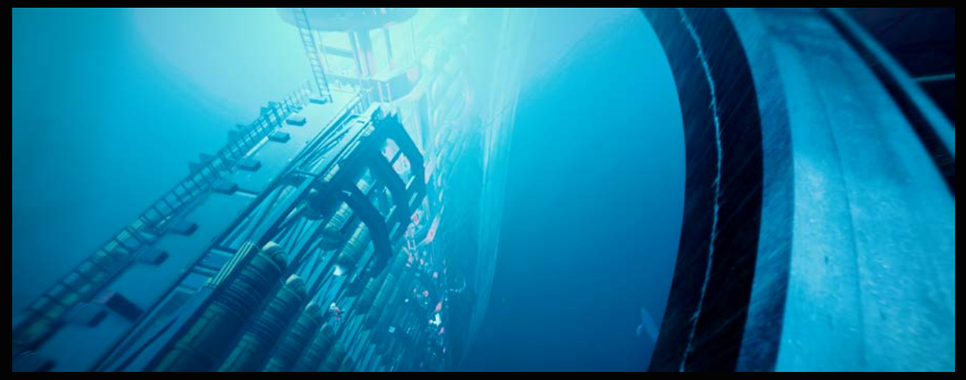










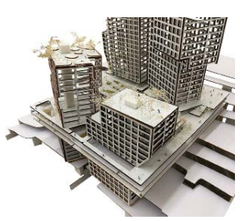


The Plat - Residential Building Design

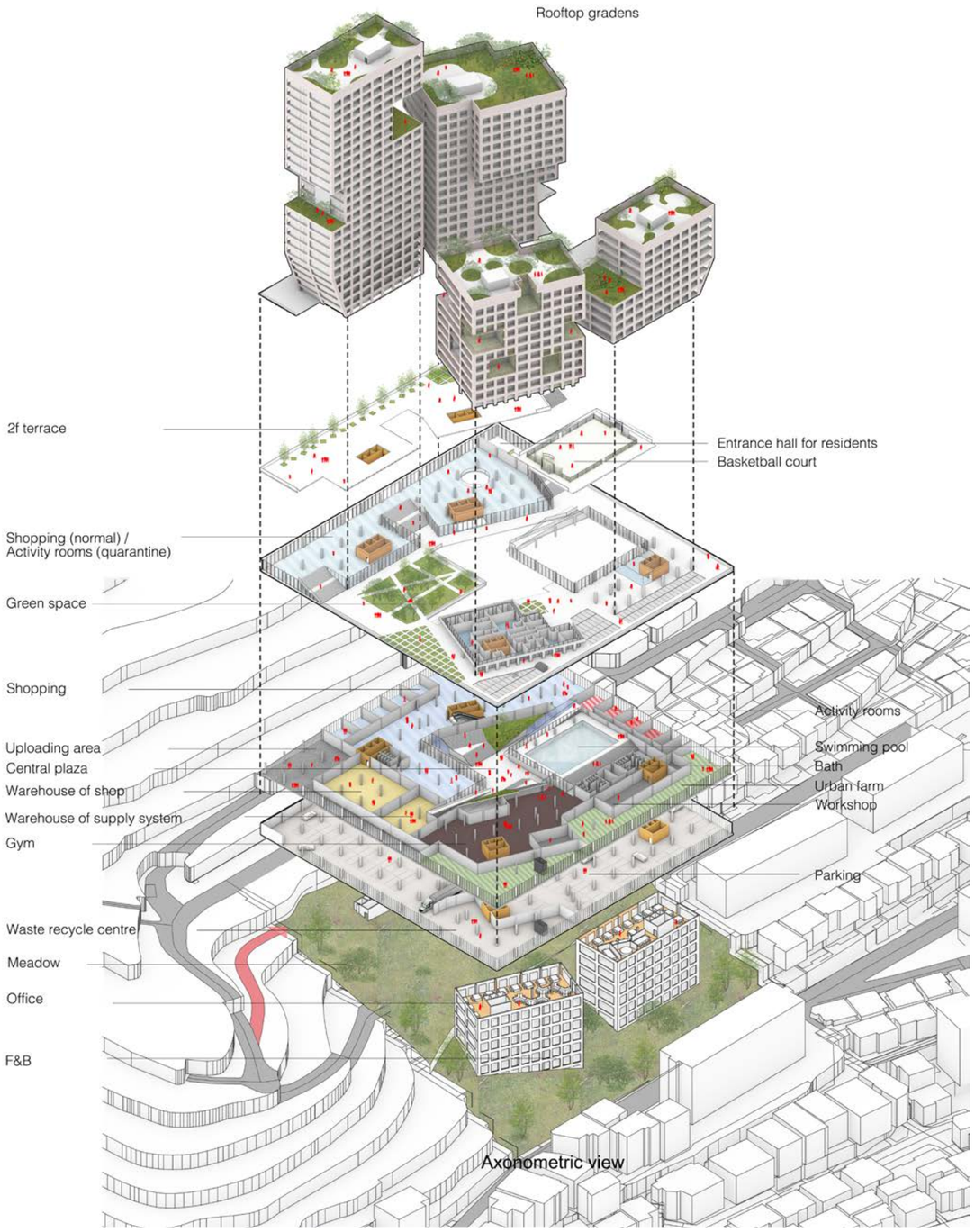






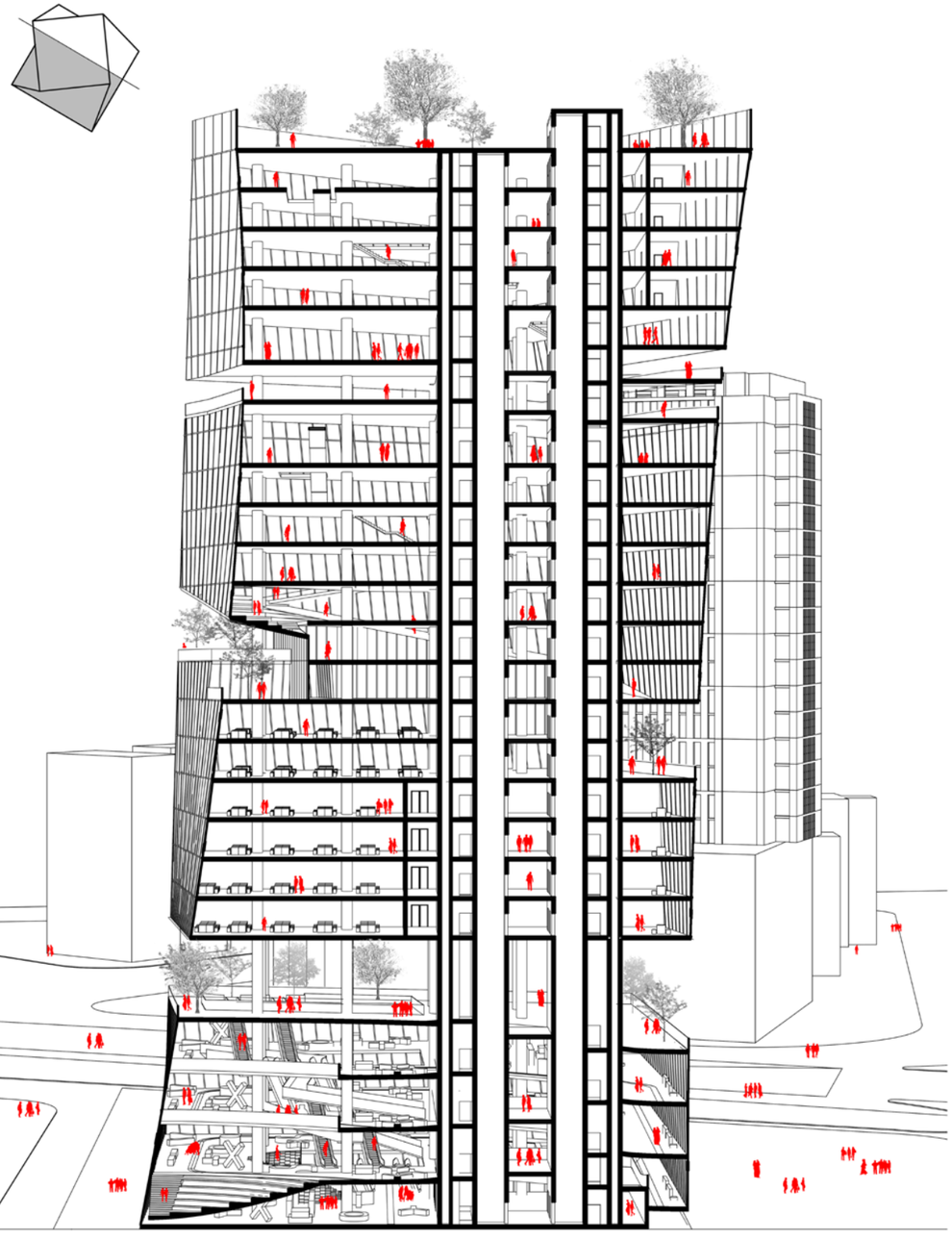


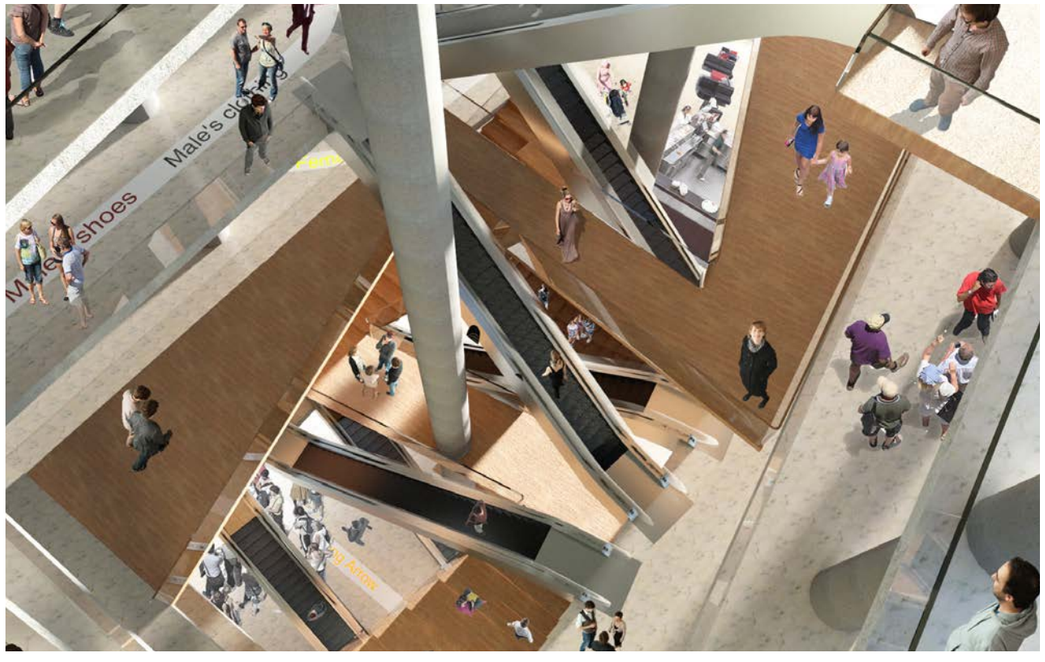


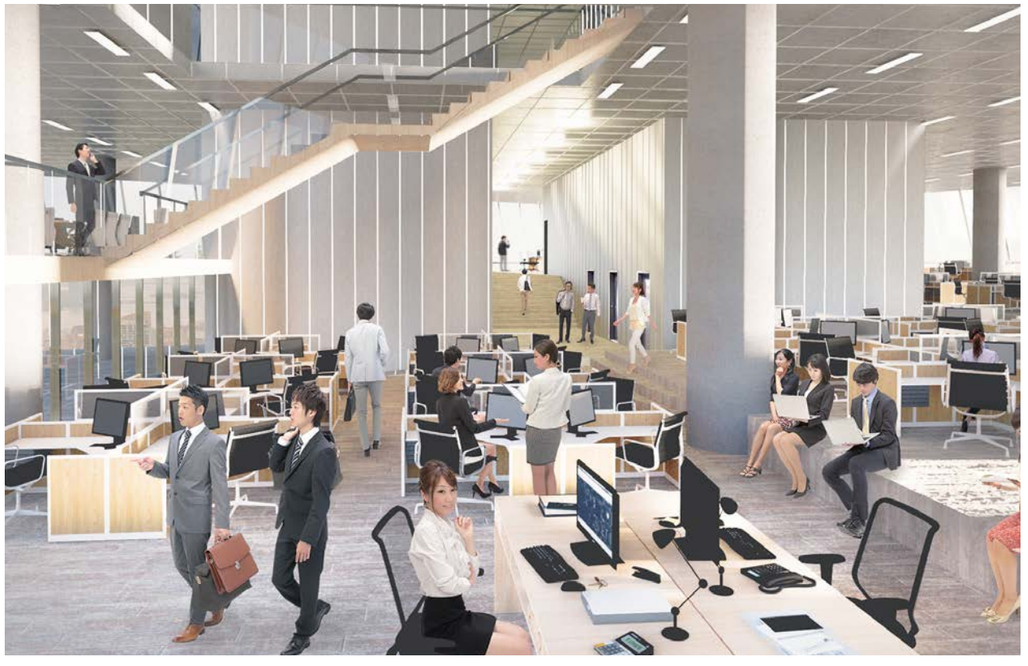


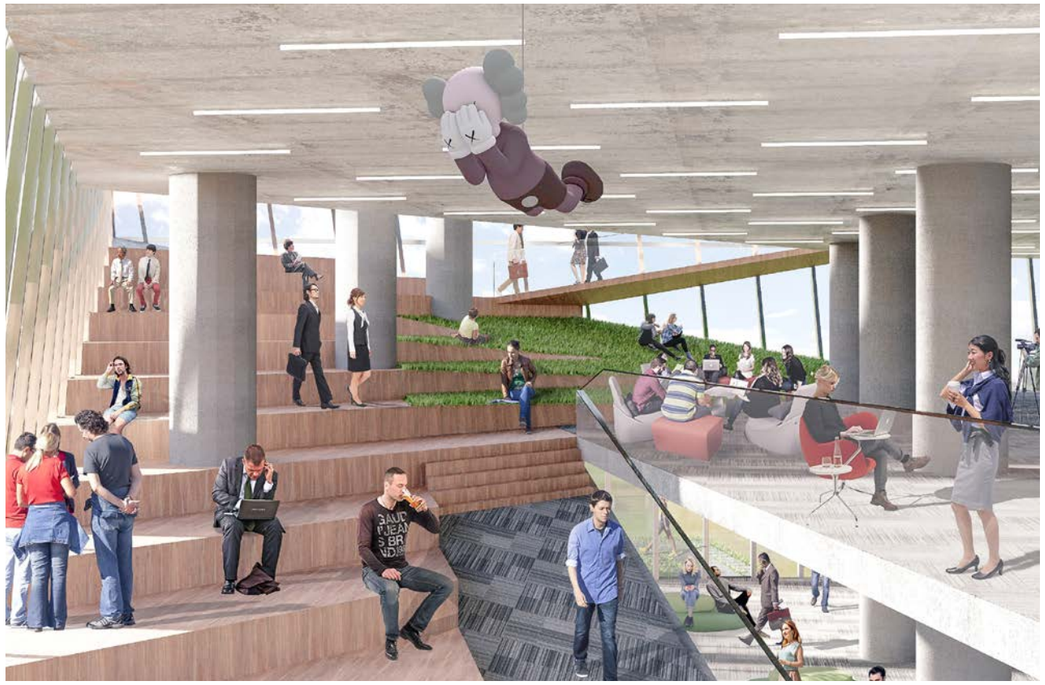
The Twist Tower





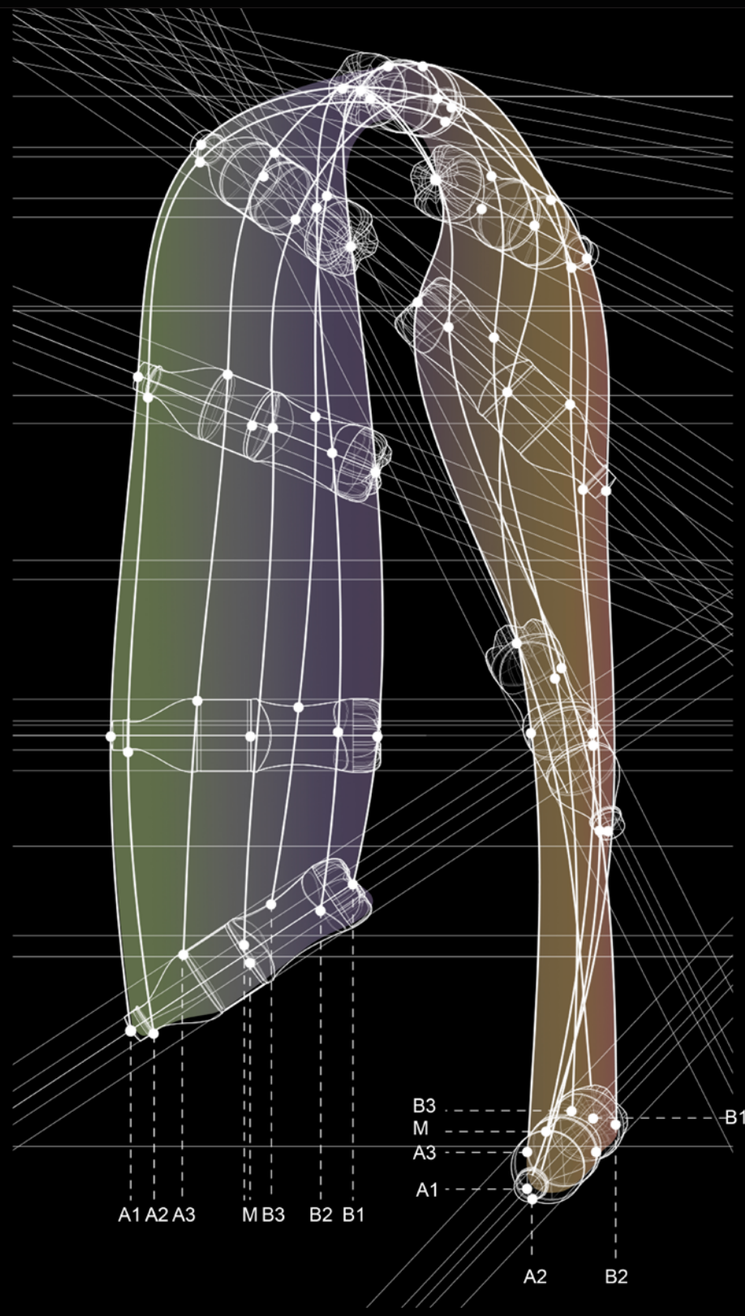

Male's shoes



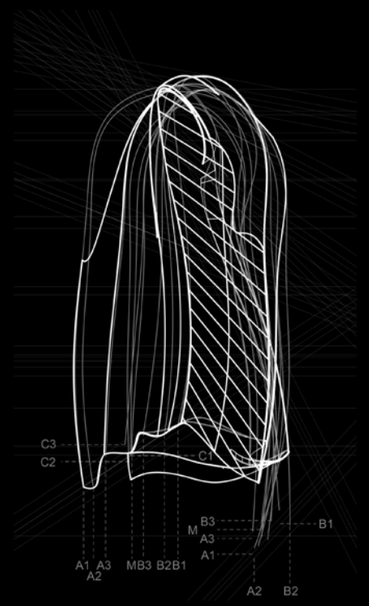




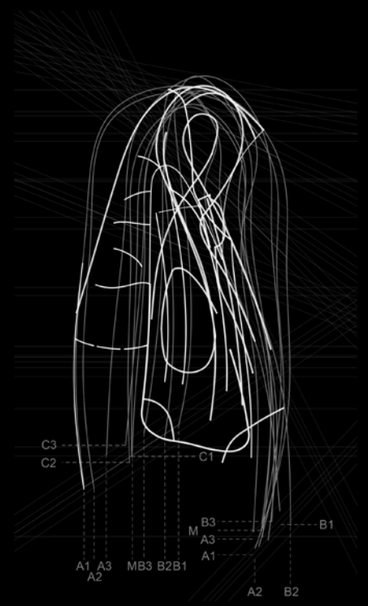
The Coke Bottle Gallery



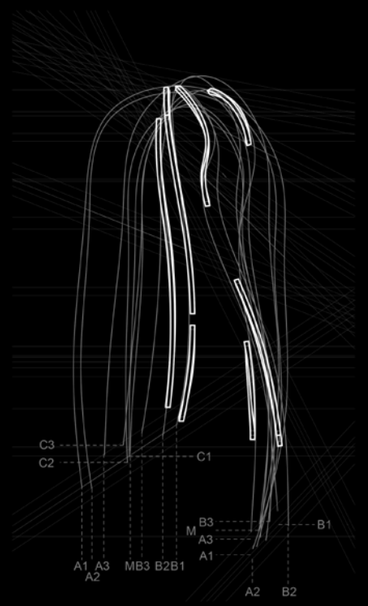
1 A few points (A1, A2, A3, M, B3, B2, B1) are captured and their movement trace are defined to form the shape of the elements of architecture.



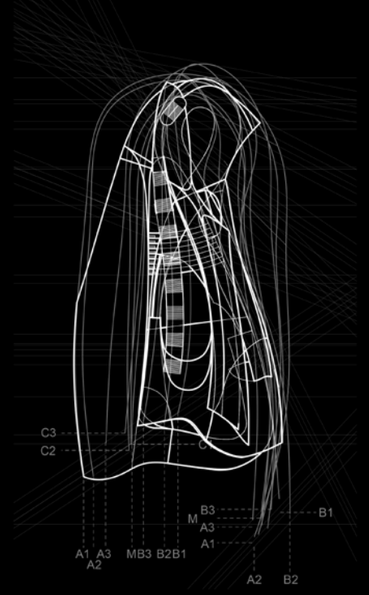
Facade



Divider

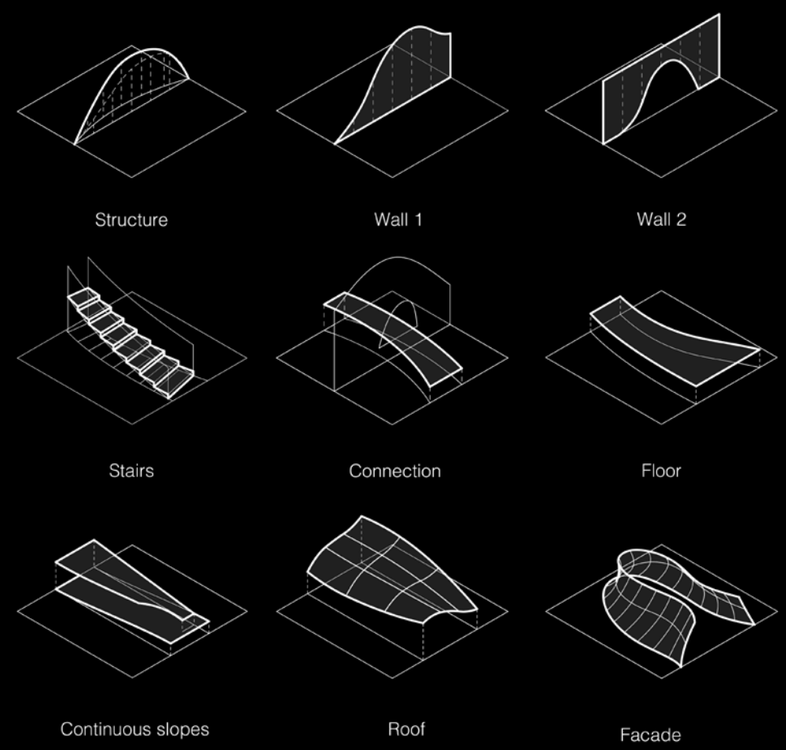


Structure



Steppers

2 The elements of architecture is generalized in four - facade (with roof), structure, divider (wall, railing), stepper (floor, slope, stairs etc.).

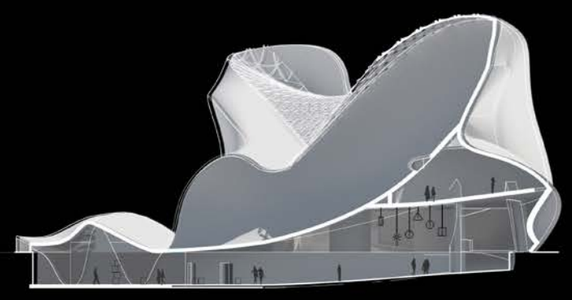


3 The planar graphs generate a series of 3D components.

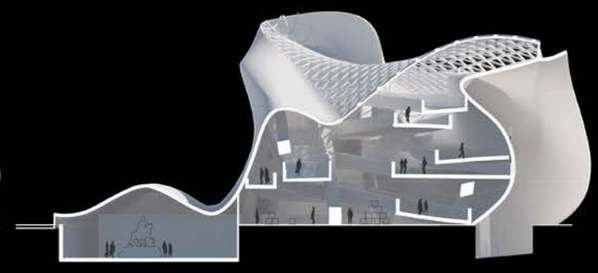




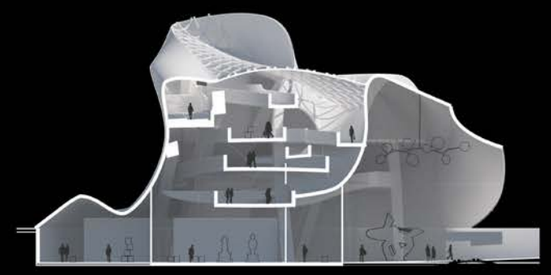




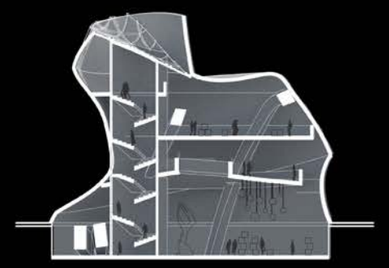
Section A-A



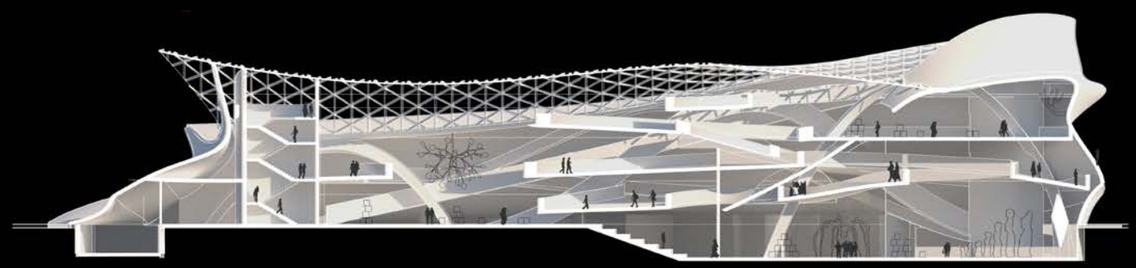
Section B-B



Section C-C



Section D-D



Section E-E